

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
OPEN TEST – I
PAPER –1
TEST DATE: 23-01-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 – 04, 20 – 23, 39 – 42): This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
Section – A (05 – 10, 24 – 29, 43 – 48): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

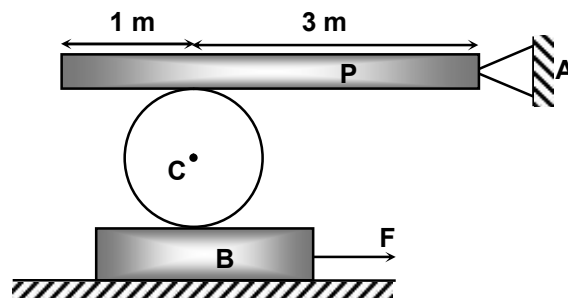
Physics

PART – I

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

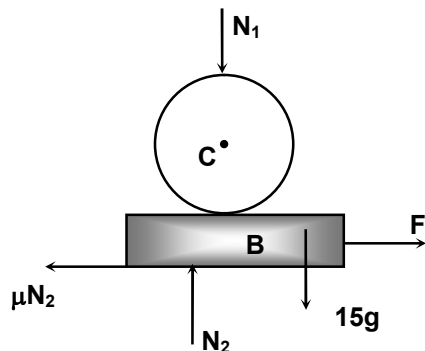
1. In the shown figure, a plank P of mass 25 kg is placed over a cylinder C of mass 10 kg. The cylinder is placed over a block B of mass 5 kg. All the surfaces are rough and coefficient of friction between all the surfaces is $\mu = 0.3$. What is the minimum horizontal force F needed to pull the block. Plank is hinged at A so that it is free to move in vertical plane without friction.



- (A) 120 N
(B) 95 N
(C) 15 N
(D) 175 N

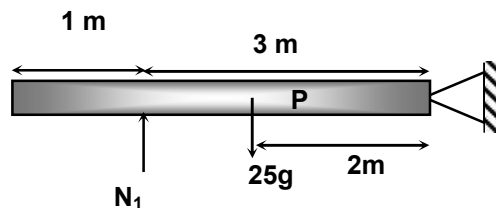
Ans. B

Sol.



$$N_2 = 15g + \frac{50}{3}g$$

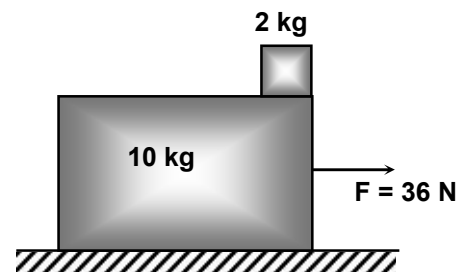
$$F = \mu N_2 = 95\text{ N}$$



$$3N_1 = 2 \times 25g$$

$$\Rightarrow N_1 = \frac{50g}{3}$$

2. A cube of side length 1 m and mass 10 kg is placed on a smooth horizontal surface. A small block of mass 2 kg and negligible size is placed on the front corner of the cube. Coefficient of friction between the cube and the block is $\mu = 0.2$. The cube is pulled with a constant horizontal force $F = 36\text{ N}$. The kinetic energy of the block just before it reaches the ground is



(A) $\frac{90}{3} \text{ J}$

(B) $\frac{80}{3} \text{ J}$

(C) $\frac{20}{3} \text{ J}$

(D) $\frac{7.2}{3} \text{ J}$

Ans. B

Sol. $a_C = \frac{36 - 0.2 \times 2g}{10} = 3.2 \text{ m/s}^2$

$a_B = \frac{0.2 \times 2g}{2} = 2 \text{ m/s}^2$

$\therefore \text{relative acceleration } a_r = 3.2 - 2 = 1.2 \text{ m/s}^2$

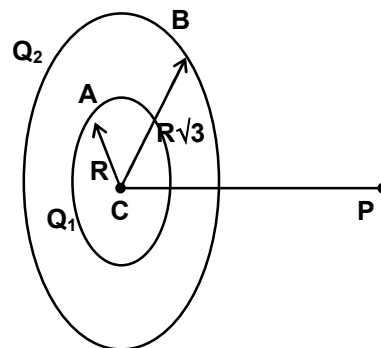
$\ell_r = \frac{1}{2} a_r t_1^2 \Rightarrow t_1 = \sqrt{\frac{2}{1.2}}$

For block $v_x = 2t_1$

Also for vertical motion $1 = \frac{1}{2} \times 10 t_2^2 \Rightarrow t_2 = \sqrt{\frac{1}{5}}$

$v_y = 10t_2, \quad KE = m(v_x^2 + v_y^2)/2$

3. In the shown figure two uniformly charged rings A and B of radius R and $R\sqrt{3}$ having charges Q_1 and Q_2 respectively are concentric with center C and coplanar. A point P is on the axis of the rings at a distance $CP = R$. At point P electric field is zero. The ratio $\left| \frac{Q_1}{Q_2} \right|$ is



(A) $\sqrt{2}$

(B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{2\sqrt{2}}$

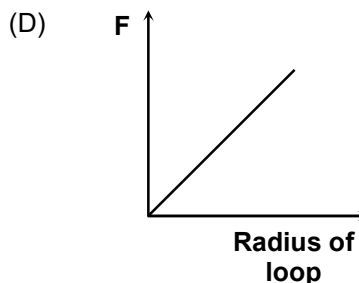
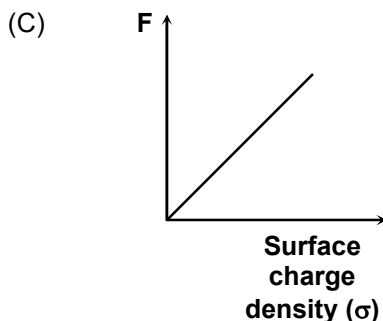
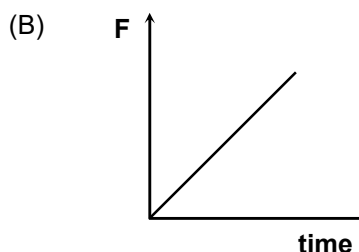
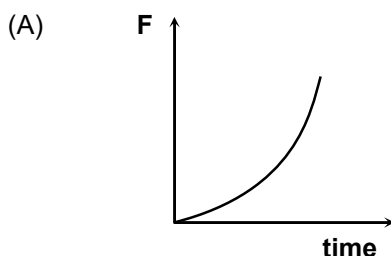
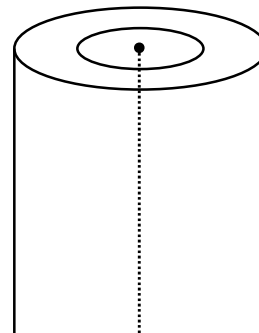
(D) $\sqrt{3}$

Ans. C

Sol.
$$E = \frac{kQ_1 R}{(\sqrt{R^2 + R^2})^3} + \frac{kQ_2 R}{(\sqrt{R^2 + 3R^2})^3} = 0$$

$$\Rightarrow \frac{Q_1}{Q_2} = -\frac{1}{2\sqrt{2}}$$

4. An insulating long cylinder is uniformly charged on its lateral surface with surface charge density σ . On its flat surface, a circular and conducting loop of radius r is placed coaxially as shown in the figure. There is no friction between the cylinder and the loop. The cylinder is being rotated with a uniform angular acceleration α about its axis. The relationship between the compressive force F in the loop and time t and surface charge density is best represented by



Ans. B

Sol. Let R_s is resistance of the loop
 Field through loop $B = (\mu_0 R \sigma \omega)/2 = (\mu_0 \sigma R \alpha t)/2$
 Induced emf in the loop $e = (\pi r^2 \mu_0 \sigma R \alpha)/2$

$$\text{Current in the loop } i = \frac{e}{R_s} = \frac{\pi r^2 \mu_0 \sigma R \alpha}{2 R_s}$$

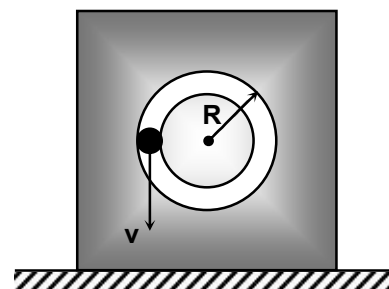
$$F = iBr = \frac{\pi r^2 \mu_0 \sigma R \alpha}{4 R_s} \mu_0 \sigma R \alpha t r$$

$$F = \left(\frac{\pi \mu_0^2 \sigma^2 \alpha^2 R^2 r^3}{4 R_s} \right) t$$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A cube of mass m and side length $4R$ has a groove made inside it in a vertical plane. The groove is a circle having radius R . A particle of mass m is moving down with a velocity $v = \sqrt{2gR}$. Assume all surfaces are smooth. The correct statement(s) at this instant.



- (A) Normal reaction on particle is mg
 (B) Normal reaction on cube by the ground is mg
 (C) Acceleration of block w.r.t ground is g
 (D) Acceleration of particle with respect to ground is $g\sqrt{2}$

Ans. A, B, C, D

Sol. $N_2 - mg = 0$

$$\Rightarrow N_2 = mg$$

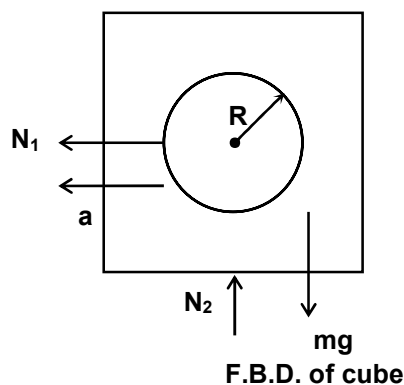
$$N_1 = ma$$

$$N_1 + ma = m\left(\frac{v^2}{R}\right)$$

$$N_1 + ma = m2g$$

$$\Rightarrow N_1 = mg$$

$$\text{and } a = g$$

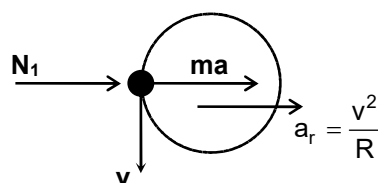


$$\vec{a}_{P,g} = \vec{a}_{P,C} + \vec{a}_{C,g}$$

$$\rightarrow g$$

$$\downarrow = g\sqrt{2}$$

$$g$$



F.B.D. of particle w.r.t.
cube

6. A black body at temperature $T = 400$ K is emitting radiations at rate 200 W/m^2 . Another body of surface area 2 m^2 coefficient of reflectivity (r) = 0.3 and zero coefficient of transmission is at temperature 800 K. The correct statement(s) about the another body is

- (A) Emissive power is 2240 W/m^2
 (B) Power emitted is 2240 W
 (C) Power emitted is 4480 W
 (D) emissivity is 0.7

Ans. A, C, D

Sol. $200 = \sigma \cdot 1 \cdot (400)^4 \quad \dots(i)$

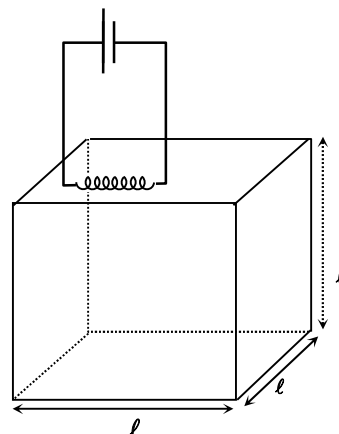
Power emitted by the other body

$P = \sigma \times 0.7 \times 2 \times (800)^4 \quad \dots(ii)$

$$\Rightarrow \frac{P}{200} = 0.7 \times 2 \times 2^4$$

$$\Rightarrow P = 200 \times 0.7 \times 2 \times 2^4$$

7. A thin cube of side length $\ell = 1\text{m}$ is heated with the help of electric heater which maintains inner temperature of cube at 100°C . The outer surface of cube emits 500 W power and absorbs 100 W radiation incident over it in steady state. After heater is switched off, it cools down according to Newton's law of cooling. When its temperature becomes 40°C at outer surface, it emits 300Watt power. Temperature of surrounding is constant at 0°C .



- (A) Power of heater is 500 W
 (B) Temperature of the outer surface is steady state 53.3°C
 (C) Temperature of outer surface in steady state 80°C
 (D) Power of heater is 400 W

Ans. C, D

Sol. Power given = power loss

$$P = 500 - 100 = 400\text{ W}$$

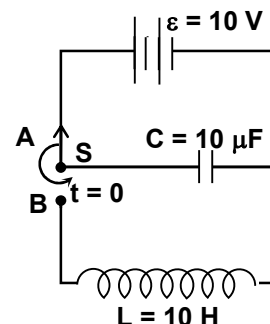
$$\frac{dQ}{dt} = k(T - 0)$$

$$400 = k(T - 0) \quad \dots(i)$$

$$\text{Also } 300 - 100 = k(40 - 0) \quad \dots(ii)$$

$$\Rightarrow T = 80^\circ\text{C}$$

8. Capacitor of capacitance $C = 10\text{ }\mu\text{F}$ was connected with a source of emf $\varepsilon = 10\text{V}$ for long time. At time $t = 0$ switch S is shifted from point A to B , which connects a solenoid of self inductance $L = 10\text{H}$ length $\ell = \frac{1}{2}\text{ m}$, and number of turns 10000 . A charge particle of charge $q = 10\text{ }\mu\text{C}$, mass $1 \times 10^{-6}\text{ kg}$ is released from rest on the axis of solenoid at time $t = \frac{10\pi}{3}\text{ ms}$. At this instant



- (A) acceleration of the charged particle is zero
 (B) acceleration of the charged particle is 100 m/s^2
 (C) magnetic field inside the solenoid is nearly 700 mT
 (D) magnetic field inside the solenoid is nearly $218 \mu\text{T}$

Ans. B, D

Sol. $V = \varepsilon_0 \cos \omega t$ is voltage across capacitor

$$\omega t = \frac{t}{\sqrt{LC}} = \frac{\frac{10\pi}{3} \times 10^{-3}}{\sqrt{10 \times 10 \times 10^{-6}}} = \frac{\pi}{3}$$

$$\therefore V = \varepsilon_0 \cos\left(\frac{\pi}{3}\right)$$

$$\Rightarrow V = \frac{10}{2} = 5 \text{ Volt}$$

$$\text{Field } E = \frac{V}{\ell} = 10 \text{ V/m}$$

$$a = \frac{qE}{m} = \frac{10 \times 10^{-6} \times 10}{1 \times 10^{-6}} = 100 \text{ m/s}^2$$

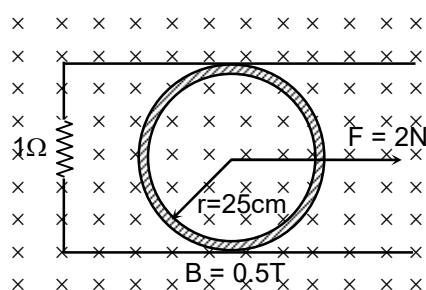
$$\frac{1}{2} C \varepsilon_0^2 = \frac{1}{2} C \left(\frac{\varepsilon_0}{2}\right)^2 + \frac{1}{2} L I^2$$

$$\Rightarrow I = 8.7 \times 10^{-3} \text{ A}$$

$$B = \mu_0 n I$$

$$= 4\pi \times 10^{-7} \times 20000 \times 8.7 \times 10^{-3} = 218 \times 10^{-6} \text{ T}$$

9. In the shown figure, two parallel rails in a vertical plane of negligible resistance are connected by a resistance $R = 1\Omega$ at one end. A uniform horizontal magnetic field is present ($B = 0.5 \text{ T}$). A thin ring of mass $m_1 = 1\text{ kg}$, radius $r = 25 \text{ cm}$ is fitted over a wooden disc of mass $m_2 = 1\text{ kg}$. Resistance of ring is $R' = 4\Omega$ and there is no friction between the ring and the disc, ring and upper rail. Friction between ring and lower rail is sufficient to support pure rolling of the ring. Horizontal force $F = 2\text{ N}$ is applied at centre of the disc. When the terminal velocity is reached



- (A) Terminal velocity of disc is 64 m/s
 (B) Terminal velocity of disc is 96 m/s
 (C) Current flowing in the rails is $8\sqrt{2} \text{ A}$
 (D) Current flowing in the rails is 8 A

Ans. A, D

Sol. at $v = v_t$

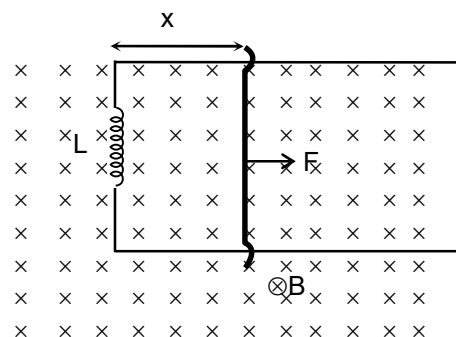
Power given = power loss

$$Fv_{\tau} = I^2 \left(1 + \frac{4}{4} \right) \quad \dots(i)$$

$$I = \frac{Bv_{\tau}(2r)}{\left(1 + \frac{4}{4} \right)} \quad \dots(ii)$$

By solving $v_{\tau} = 64 \text{ m/s}$
and $I = 8 \text{ A}$

10. A conducting wire of mass m is pulled with a uniform velocity v_0 on two parallel rails lying in horizontal plane and separated by distance ℓ . Ends of rails are connected by a coil of self inductance L . At time $t = 0$, $x = x_0 (\approx 0)$ and uniform vertical magnetic field B is present in space. When the conducting wire is at distance x from the initial position.



- (A) Force F is directly proportional to x
 (B) Force F is directly proportional to x^2
 (C) Energy stored in coil is directly proportional to x
 (D) Energy stored in coil is directly proportional to x^2

Ans. A, D

Sol.
$$L \frac{dI}{dt} = B\ell \frac{dx}{dt}$$

$$\Rightarrow \int_0^I dI = \frac{B\ell}{L} \int_0^x dx$$

$$I = \frac{B\ell}{L} x$$

$$U = \frac{1}{2} LI^2 \Rightarrow U \propto x^2$$

$$F = IB\ell$$

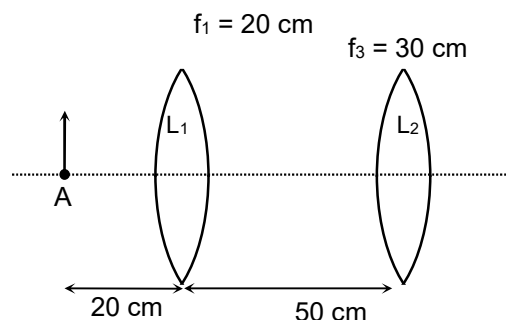
$$F = \frac{B^2 \ell^2}{L} x$$

$$F \propto x$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. Two converging lenses L_1 and L_2 of focal lengths $f_1 = 20$ cm and $f_2 = 30$ cm respectively are placed such that their principal axes coincide. An object is first placed at a distance 25 cm from lens L_1 . Object is oscillating along principal axis with amplitude 5 cm. Find the ratio of maximum to minimum magnification of its image.



Ans. 1

Sol. Magnification is independent of object distance from lens L_1 .
Since separation between lenses $d = f_1 + f_2$

12. Two blocks of equal masses are placed one over the other as shown in the figures. Friction between all the contact surfaces is $\mu = 0.8$. Minimum values of the forces required to pull the blocks are F_1 and F_2 respectively. Force F_1 is horizontal and F_2 is parallel to the inclined plane.

Find the ratio $\frac{F_1}{F_2}$.

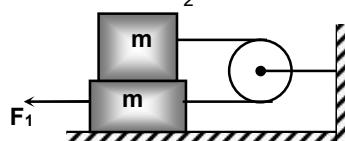


Figure-A

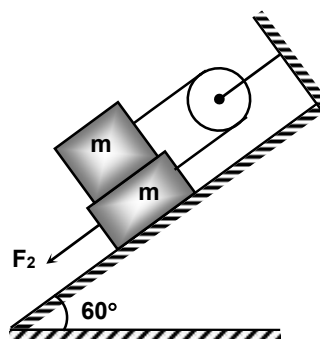
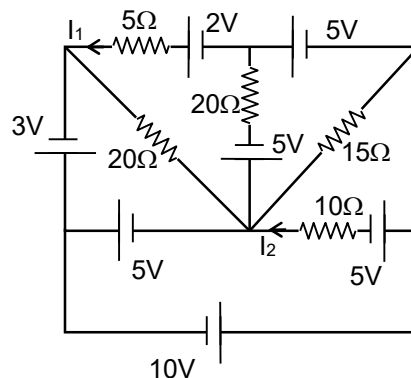


Figure-B

Ans. 2

Sol. $F = T + 2\mu mg + \mu mg$
And $T = \mu mg$
 $\therefore F_1 = 4\mu mg$
 $F_2 + mg \sin 60^\circ = T + 3\mu mg \cos 60^\circ$
 $F_T = mg \sin 60^\circ + \mu mg \cos 60^\circ$
 $\therefore F_2 = 4\mu mg \cos 60^\circ$
 $\frac{F_1}{F_2} = \frac{1}{\cos 60^\circ} = 2$

13. In the shown circuit all the sources of emf are ideal. Current through 5Ω resistance is I_1 and through 10Ω resistance is I_2 . Find the ratio $\left(\frac{I_1}{I_2}\right)$.



Ans. 4

Sol. Using Kirchoff's law
 $-3 + 5I_1 - 2 - 5 - 10 = 0 \Rightarrow I_1 = 4 \text{ A}$
 and $-5 + 10I_2 + 5 - 10 = 0 \Rightarrow I_2 = 1 \text{ A}$
 $\Rightarrow \frac{I_1}{I_2} = 4$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

A block of ice at 0°C and mass M is kept inside isolated container of negligible heat capacity. Steam of mass m at temperature 100°C is kept inside the container. The ice melted and its temperature becomes 20°C . The steam of mass 0.2 kg at 100°C is again let inside the container, due to which temperature inside the container becomes 30°C in steady state. Latent heat of ice and steam are 80 kcal/kg and 540 kcal/kg respectively. Specific heat capacity of water is $S_w = 1 \text{ kcal/kg}^\circ\text{C}$.

14. The value of m (in kg) is

Ans. 1.69

15. The value of M (in kg) is

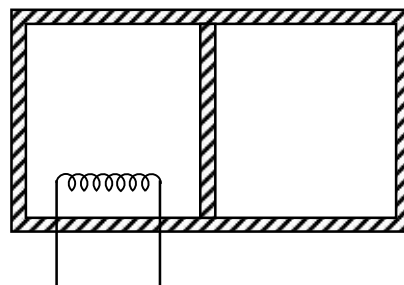
Ans. 10.48

Sol. (for Q.14-15)
 Using conservation of energy
 $(m \times 540) + m \times 1 \times (100 - 20) = (M \times 80) + M \times 1 \times (20 - 0)$
 $\Rightarrow M = 6.2 \text{ m} \quad \dots(i)$
 $(0.2 \times 540) + 0.2 \times 1 \times (100 - 30) = (M + m) \times 1 \times (30 - 20)$
 $\Rightarrow M + m = 12.2$
 Solving equations
 $m = 1.69 \text{ kg}$
 $M = 10.48 \text{ kg}$

Question Stem for Question Nos. 16 and 17

Question Stem

An insulating cylinder of volume 20 liter is divided into two equal parts by a movable piston made of insulating material. Monoatomic ideal gas is filled at temperature 300 K and pressure 10^5 N/m^2 in each part. Left part is given heat Q joule by using an electric heater such that temperature in the right part becomes $(300 + 2) \text{ K}$ and in the left part becomes $(300 + \Delta T) \text{ K}$. (Take $\left(\frac{300}{302}\right)^{3/2} = 0.99$)



16. The value of ΔT is

Ans. 8.10

17. The value of Q is

Ans. 50.50

Sol. (for Q. 16-17)

In right part using adiabatic equation

$$300 \times 10^{2/3} = (300 + 2) V_2^{2/3}$$

$$\Rightarrow V_2 = 9.9 \text{ liter}$$

Pressure is same in left part also temperature in left part T'_2

Volume in left part $V'_2 = 20 - 9.9 = 10.1 \text{ liter}$

$$T'_2 = \left(\frac{10.1 \times 302}{9.9} \right) = 308.10,$$

By solving $T'_2 = 308.10$

$$\therefore \Delta T = 308.10 - 300 = 8.10$$

$$Q = nC_v \Delta T_1 + nC_v \Delta T_2$$

$$nC_v (\Delta T_1 + \Delta T_2)$$

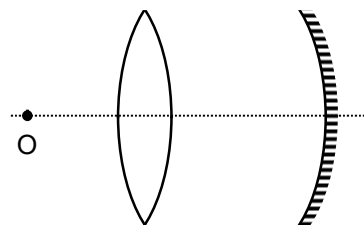
$$\frac{P_1 V_1}{RT_1} \times \frac{3}{2} R (\Delta T_1 + \Delta T_2) = \frac{3}{2} \frac{P_1 V_1}{T_1} (\Delta T_1 + \Delta T_2)$$

$$= \frac{3}{2} \times \frac{10^5 \times (10 \times 10^{-3})}{300} (2 + 8.10) = 50.50 \text{ J}$$

Question Stem for Question Nos. 18 and 19

Question Stem

A converging lens of power $P_1 = +5D$ and a concave mirror of power $P_2 = +16D$ are placed at a separation 80 cm such that their principal axis coincide. A point object is placed left side of the lens on its principal axes at a distance X_1 from the lens such that image of the object coincides with the object itself. When the mirror is further shifted right side by a distance X_2 , the image of the object further coincides with the object.



18. The value of X_1 (in cm) is

Ans. 26.67

19. The value of X_2 (in cm) is

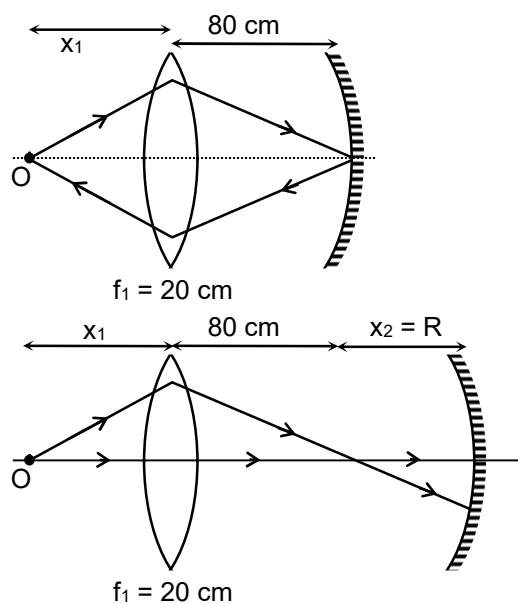
Ans. 12.50

Sol. (for Q.18-19)

$$\frac{1}{80} - \frac{1}{-x_1} = \frac{1}{20}$$

$$\frac{1}{x_1} = \frac{1}{20} - \frac{1}{80} \Rightarrow x_1 = 26.67 \text{ cm}$$

$$x_2 = R = 12.50 \text{ cm}$$



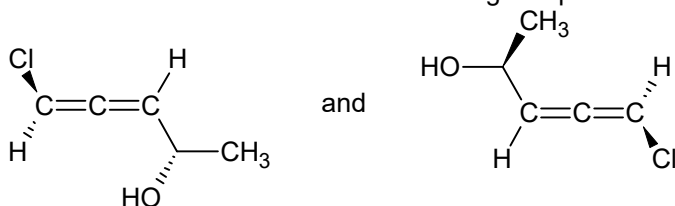
Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. The correct relation between the following compounds is:

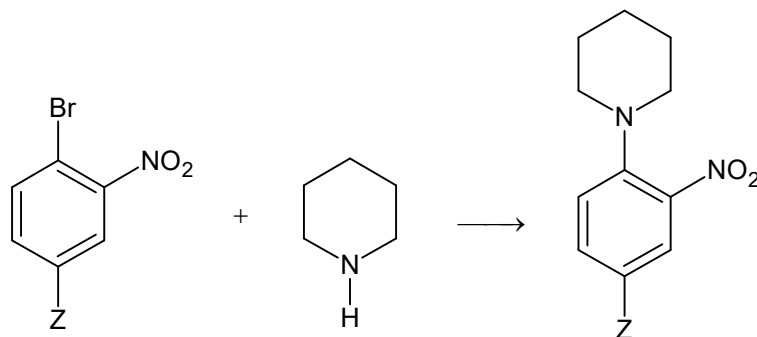


- (A) Enantiomers
(B) Diastereomers
(C) Homomers (identical)
(D) Constitutional isomers

Ans. C

Sol. Since, the absolute configuration about the chiral carbon is (S) in each compound. Also, configuration of Cl and H about double bond carbon is also same in each compound. So, both compounds are identical.

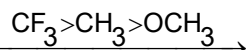
21. The correct order of the rate constants for the following series of reactions ($Z = \text{CF}_3, \text{CH}_3, \text{OCH}_3$):



- (A) $\text{CF}_3 > \text{CH}_3 > \text{OCH}_3$
(B) $\text{CF}_3 > \text{OCH}_3 > \text{CH}_3$
(C) $\text{OCH}_3 > \text{CF}_3 > \text{CH}_3$
(D) $\text{CH}_3 > \text{OCH}_3 > \text{CF}_3$

Ans. A

Sol. Rate of $\text{ArS}_\text{N}2 \propto$ magnitude of positive charge on α – carbon atom.



Decreasing electron withdrawing nature and hence decreasing rate of $\text{ArS}_\text{N}2$.

22. The left half-cell of the electrochemical cell consists of an iron electrode (in excess) that is oxidized during the action and an aqueous solution of iron (III) nitrate. The right half-cell consists of graphite electrode and a mixture of iron (II) and (III) nitrates. Using the information on ion mobility (μ°), select the most suitable ionic compound (s) for the salt bridge:

Cation	$\mu^\circ, \text{m}^2 \text{V}^{-1} \text{s}^{-1} \cdot 10^9, 25^\circ \text{C}$	Anion	$\mu^\circ, \text{m}^2 \text{V}^{-1} \text{s}^{-1} \cdot 10^9, 25^\circ \text{C}$
$\text{Ca}_{(\text{aq})}^{2+}$	61.6	$\text{F}_{(\text{aq})}^-$	51.5
$\text{K}_{(\text{aq})}^+$	73.5	$\text{CO}_3^{2-}(\text{aq})$	70.3
$\text{NH}_4^+(\text{aq})$	76.1	$\text{NO}_3^-(\text{aq})$	74.1
$\text{H}_{(\text{aq})}^+$	205	$\text{Br}^-(\text{aq})$	81.8

- (A) Calcium carbonate
 (B) Ammonium nitrate
 (C) Potassium carbonate
 (D) Ammonium bromide

Ans. B

Sol. For the salt bridges, the most advantageous is to use salts in which the cation and the anion have the same mobility which assure the absence of so called diffusion potential in the half-cell boundary. Thus, the most suitable salts are KNO_3 and NH_4NO_3 .

23. 50 mL of 0.10 M HX has been mixed with 100 mL of 0.05 M NaOH, then the concentration of the conjugate base of HX in the resultant solution can't be
- (A) 0.032 M
 (B) 0.028 M
 (C) 0.036 M
 (D) Both (A) and (B)

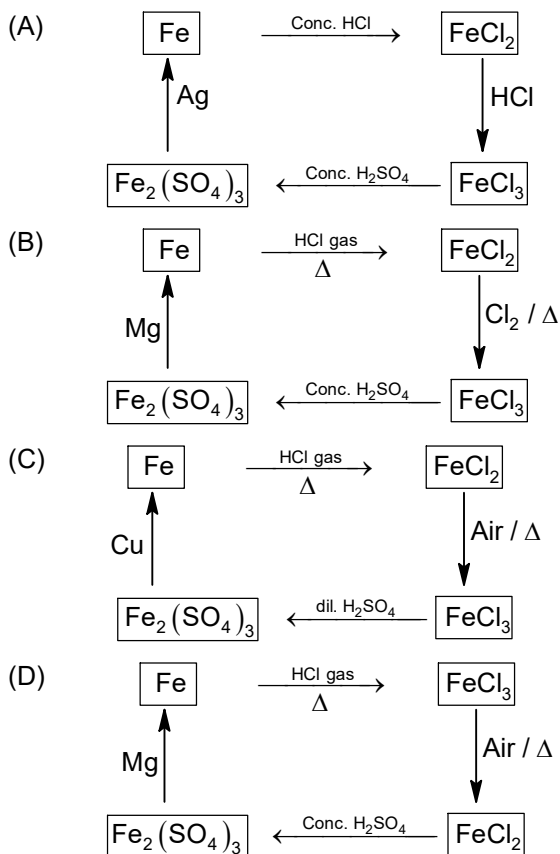
Ans. C

Sol. If HX is a strong acid then maximum concentration of conjugate base is 0.033 M. Therefore, any value higher than this is not possible.

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

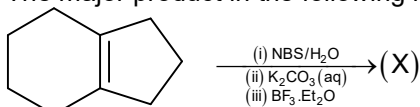
24. Which systematic diagram represents the “INCORRECT” chemical relations between iron and its compounds?



Ans. A, C, D

Sol. Iron reacts with dry HCl gas to give FeCl_2 . FeCl_2 on heating with chlorine gas (dry) oxidises to FeCl_3 . Conc. H_2SO_4 being less volatile substitute more volatile HCl from FeCl_3 , converting FeCl_3 into $\text{Fe}_2(\text{SO}_4)_3$. Mg being more electropositive than iron reduces $\text{Fe}_2(\text{SO}_4)_3$ into Fe.

25. The major product in the following reaction is (X).

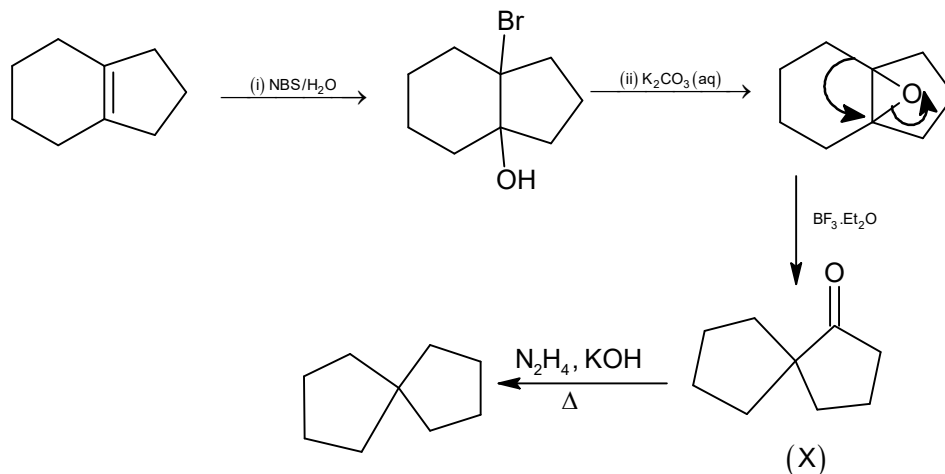


Identify the correct statement(s) about (X):

- (A) 'X' gives positive Brady's test
 (B) 'X' gives positive Bromine-water test
 (C) 'X' on reduction with $\text{N}_2\text{H}_4 / \text{KOH} / \Delta$, gives a bicyclic compound
 (D) 'X' is a bicyclic compound and both rings in 'X' are of same size

Ans. A, C, D

Sol.



26. On adding potassium iodide to an aqueous solution containing complex ion $[\text{Ag}(\text{NH}_3)_2]^+$, yellow coloured precipitate of AgI is formed. Similarly, when hydroiodic acid was added to the complex ion, $[\text{Ag}(\text{NH}_3)_2]^+$ again yellow precipitate of AgI is formed. The correct statement(s) for these observations is/are:

- (A) $[\text{Ag}(\text{NH}_3)_2]^+$ is less stable than AgI.
 (B) $[\text{Ag}(\text{NH}_3)_2]^+$ ionizes less than AgI.
 (C) HI neutralizes NH_3 ligands and precipitates AgI.
 (D) IUPAC name of $[\text{Ag}(\text{NH}_3)_2]^+$ is "Diamminesilver (I) ion".

Ans. A, C, D

Sol. AgI is more stable than $[\text{Ag}(\text{NH}_3)_2]^+$. HI neutralises NH_3 for forming NH_4^+ ion, thus it cannot act as ligand.

27. Which of the following statement(s) is/are correct?

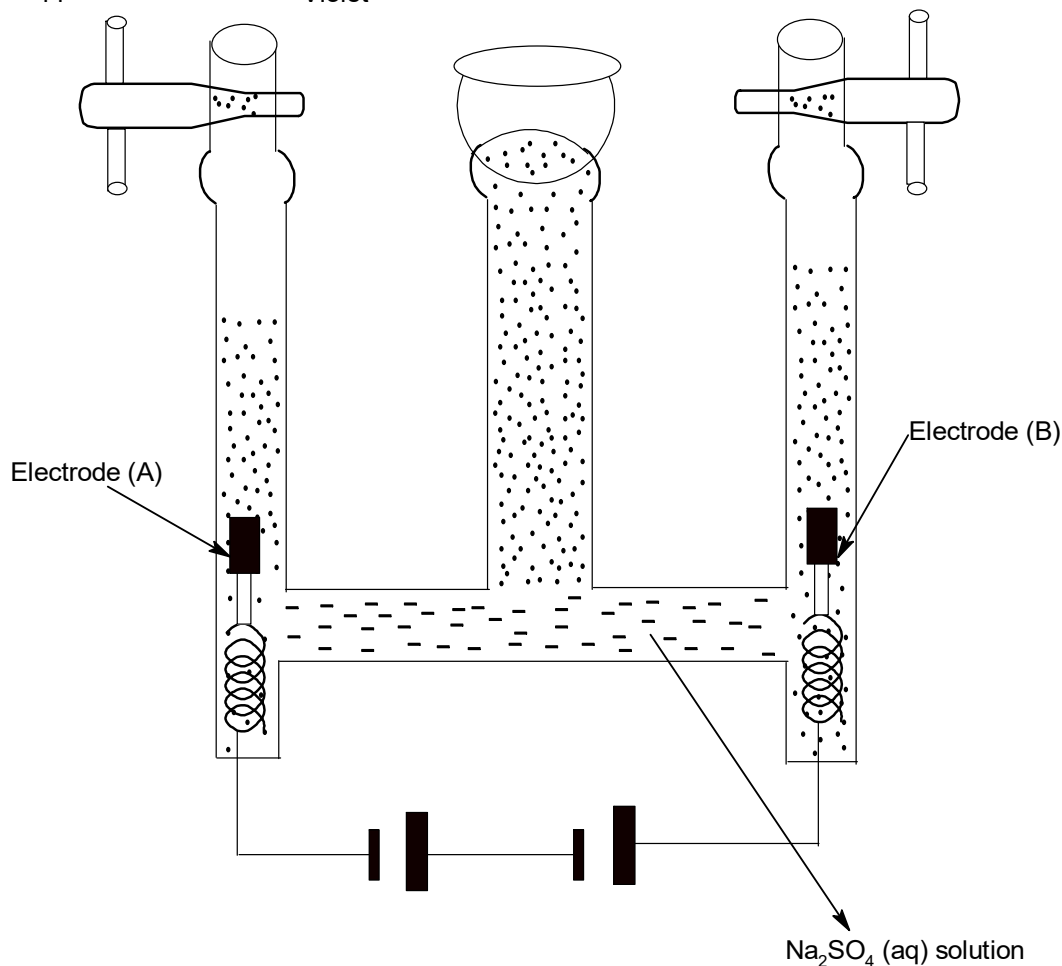
- (A) Cu^{2+} salts form soluble complex with excess KCN.
 (B) Cu^{2+} salts form soluble complex with aqueous ammonia.
 (C) Cu^{2+} salts form soluble complex with KI.
 (D) A piece of iron or zinc when placed in Cu^{2+} salts solution, precipitates copper.

Ans. A, B, D

- Sol. (A) $2\text{Cu}^{2+} + 10\text{KCN} \longrightarrow 2\text{K}_3[\text{Cu}(\text{CN})_4] + (\text{CN})_2 \uparrow + 4\text{K}^+$
soluble complex
- (B) $\text{Cu}^{2+} + 4\text{NH}_3(\text{aq}) \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$
(soluble blue complex)
- (C) $2\text{Cu}^{2+} + 4\text{KI} \longrightarrow 2\text{CuI} \downarrow + \text{I}_2 + 4\text{K}^+$
(ppt.)
- (D) In electrochemical series, Fe and Zn both are above Cu. So, both Fe and Zn can displace Cu^{2+} from its salt into Cu.

28. The apparatus shown was used to investigate the electrolysis of aqueous sodium sulphate solution to which a few drops of universal indicator solution had been added. The solution was initially pale green in colour. Colour of universal indicator at different pH values is given below:

pH range	Colour
< 3	Red
3 – 6	Orange-yellow
7	Green
8 – 11	Blue
> 11	Violet



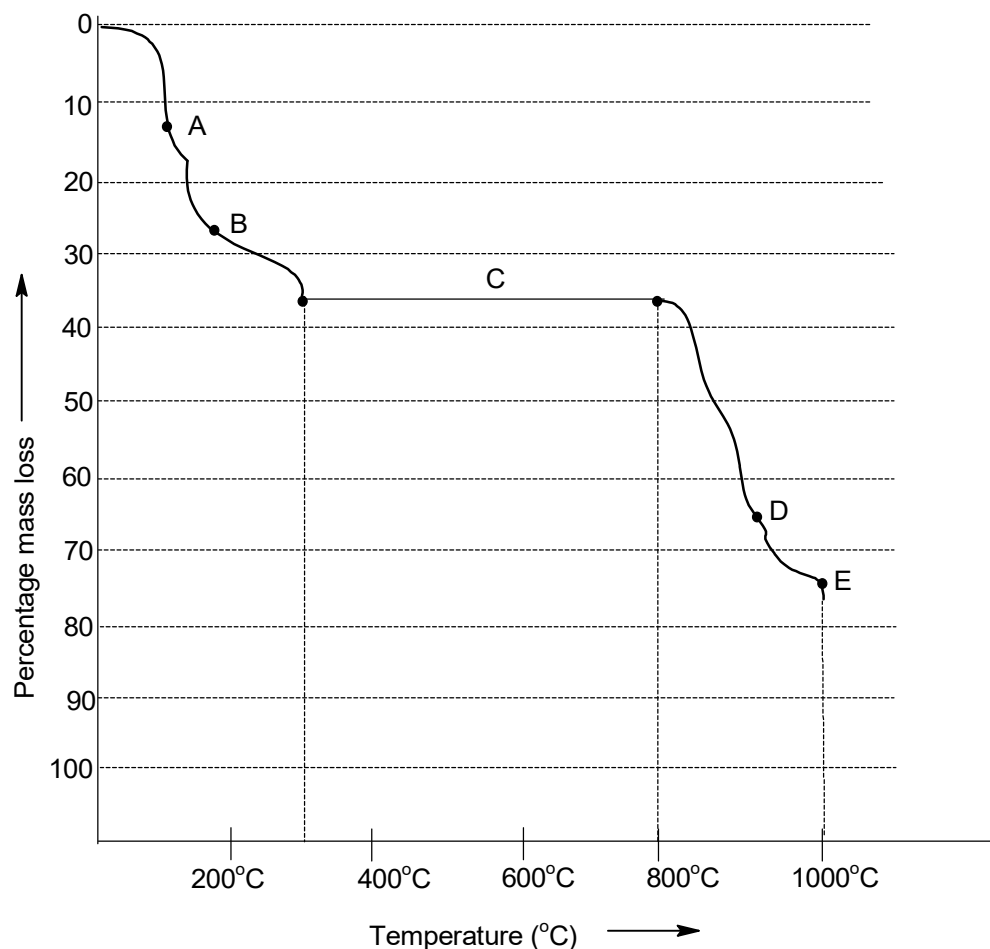
Now, select the correct statement(s) among the following:

- (A) Current conducted through the electrolyte is due to the movement of ions.
- (B) Oxidation occurs at electrode 'B'.
- (C) When a current was passed through the electrolyte for 15 minutes, 8.5 cm^3 of gas was collected above the positive electrode, then the volume of the gas collected above the negative electrode at same pressure and temperature and in the same duration is 25 cm^3 .
- (D) Blue colour appears near the electrode 'A'.

Ans. A, B, D

Sol. Milli-equivalent of H_2 gas produced at cathode are same as milli-equivalents of O_2 produced at anode. As the solution around electrode 'A' is slightly basic, colour changes to blue.

29. The graph below shows the change in the mass as copper (II) sulphate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) when heated. Decomposition takes place where the gradient is steepest leaving various decomposition products indicated by 'A' to 'E' in the diagram shown below: (Atomic mass: Cu = 63.5 u, S = 32 u, O = 16 u, H = 1 u)

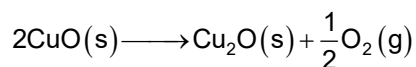
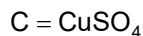
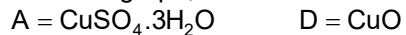


Select the correct statement (s) among the following:

- (A) Conversion of (D) to (E) is a redox reaction with n-factor of one of the product as 4.
- (B) (C) is a white compound.
- (C) Ratio of oxygen atoms per formula unit in (A) to (B) is 3.
- (D) Copper (II) sulphate decomposes above 800°C .

Ans. A, B, D

Sol. From the graph, it can be concluded that



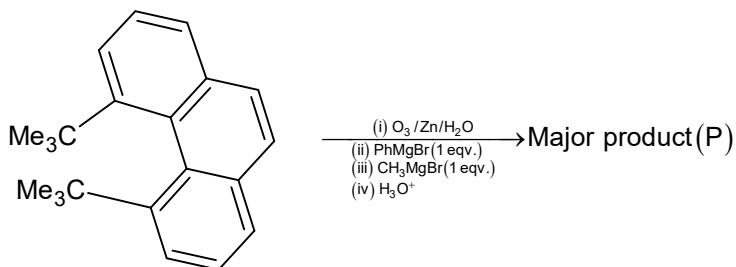
O_2 has n-factor 4.

About 35% loss in mass corresponds to loss of all water molecules.

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. Consider the following reaction



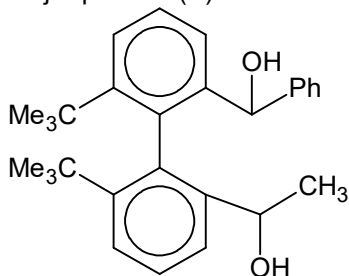
Let, degree of unsaturation of (P) = x .

Possible number of stereoisomers of (P) = y

So, the value of $(x + y)$ is _____

Ans. 20

Sol. Major product (P) is



So, $x = 12$, and $y = 8$.

31. How many of the following statement(s) regarding oxides (M_2O), peroxides (M_2O_2) and superoxide (MO_2) of alkali metals are correct?
- (i) Order of size of oxide, peroxide and superoxide ions is $O^{2-} < O_2^{2-} < O_2^-$.
 - (ii) Thermal stability order is $O^{2-} < O_2^{2-} < O_2^-$.
 - (iii) Thermal stability order with a particular alkali metal is : superoxide < peroxide < oxide.
 - (iv) Lattice energy of oxides, peroxides and superoxides decreases down the group for alkali Metals.
 - (v) In the formation of oxides, peroxides and superoxides, the entropy decreases in the order : oxide < peroxide < superoxide.
 - (vi) Heat of formation is in the order : oxide > peroxide > superoxide.
 - (vii) Heat of formation decreases down the group for oxides, peroxides and superoxides.
 - (viii) Compared to the difference in lattice energies between monoxides and peroxides is less than between monoxides and superoxides.
 - (ix) Compared to the difference in lattice energies between monoxides and peroxides is less than between peroxides and superoxides.
 - (x) All oxides, peroxides and superoxides form basic solutions when dissolved in water.
 - (xi) Thermal stability of oxides and peroxides decreases down the group while those of superoxides increases down the group.
 - (xii) The difference between the lattice enthalpies of monoxides and peroxides decreases down the group.

Ans. 8

Sol. Correct statements are :
(iii); (iv); (v); (viii); (ix); (x); (xi) and (xii).

Important data for support:

⇒ Lattice energies (kJ mol^{-1})

	M_2O	M_2O_2	MO_2
Li	2907	2592	878
Na	2518	2309	799
K	2229	2114	741
Rb	2146	2025	706
Cs	2016	1948	679

⇒ Standard enthalpies of formation (ΔH_f°) in kJ mol^{-1} :

	M_2O	M_2O_2	MO_2
Li	-596	-640	-
Na	-416	-505	-270
K	-362	-494	-280
Rb	-330	-426	-310
Cs	-318	-402	-315

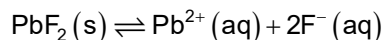
32. In a solution of 0.010 M barium nitrate and 0.010 M lead (II) nitrate, when the more soluble fluoride (other than NaF) begins to precipitate, the concentration of the cation for the less soluble fluoride that remains in solution, when NaF is added, is $a \times 10^{-b}$ M. So, the value of $(10 \times a + b)$ is.....

$$[K_{sp} \text{ of } \text{PbF}_2 = 4 \times 10^{-8} \text{ and } K_{sp}(\text{BaF}_2) = 1.44 \times 10^{-6}]$$

(Round off the answer to the nearest integer)

Ans. 32

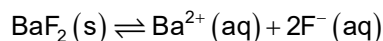
Sol. For PbF_2 ,



$$4 \times 10^{-8} = (0.01) \times [\text{F}^{-}]^2$$

$$\therefore [\text{F}^{-}] = 2 \times 10^{-3} \text{ M}$$

For BaF_2 :



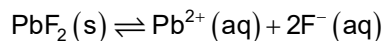
$$1.44 \times 10^{-6} = (0.01) \times [\text{F}^{-}]^2$$

$$[\text{F}^{-}] = 1.2 \times 10^{-2} \text{ M}$$

The PbF_2 will precipitate first because a lower value for the concentration of fluoride ion is needed for its precipitation.

BaF_2 starts precipitating when the conc. Of F^{-} reaches to 1.2×10^{-2} M.

Since, $\text{PbF}_2(\text{s})$ is present then



$$4 \times 10^{-8} = (\text{Pb}^{2+}) \times (1.2 \times 10^{-2})^2$$

$$(\text{Pb}^{2+}) = \frac{4 \times 10^{-8}}{1.2 \times 1.2 \times 10^{-4}} = 2.77 \times 10^{-4} \approx 2.8 \times 10^{-4}$$

$$\therefore a = 2.8, b = 4$$

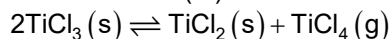
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

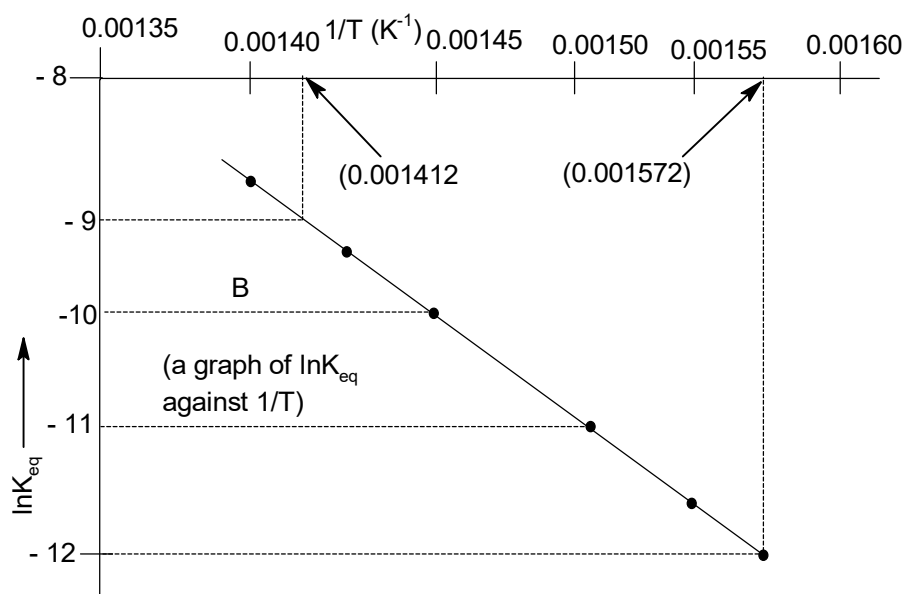
Question Stem for Question Nos. 33 and 34

Question Stem

When titanium (III) chloride is heated, it undergoes the following reaction:



A graph showing how ' K_{eq} ' varies with temperature for this reaction is plotted below



Graph not to scale

Assume standard enthalpy change (ΔH°) and standard entropy change (ΔS°) are temperature independent in the temperature range given in the graph. [$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

Let, the (ΔH°) of the reaction be ' x ' kJ and standard free energy change (ΔG°) of the reaction at 689.65 K be ' y ' (kJ).

33. The value of ' x ' is _____

Ans. Range (155.85 – 155.90)

Sol. $\ln K_{\text{eq}} = \frac{-\Delta H^\circ}{R} \times \frac{1}{T} + \frac{\Delta S^\circ}{R} \quad \dots (1)$

So, the slope of $\ln K_{\text{eq}}$ against $\left(\frac{1}{T}\right) = -\frac{\Delta H^\circ}{R}$ from the graph.

$$\text{Slope} = -\frac{\Delta H^\circ}{R} = \frac{-12 - (-9)}{0.001572 - 0.001412} = -18750$$

$$\therefore -\frac{\Delta H^\circ}{R} = -18750$$

$$\text{or } \Delta H^\circ = \frac{18750 \times 8.314}{1000} = +155.88 \text{ kJ}$$

34. The value of 'y' is _____

Ans. Range (55.65 – 55.70)

Sol. $\ln K_{eq} \left(-\frac{\Delta H^\circ}{R} \right) \frac{1}{T} + \frac{\Delta S^\circ}{R}$

$$-12 = -18750 \times 0.001572 + \frac{\Delta S^\circ}{R}$$

$$\text{So, } \frac{\Delta S^\circ}{R} = 17.475$$

$$\therefore \Delta S^\circ = 17.475 \times 8.314 = 145.28 \text{ JK}^{-1}$$

$$\text{So, } \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= 155.88 \times 10^3 - 689.65 \times 145.28$$

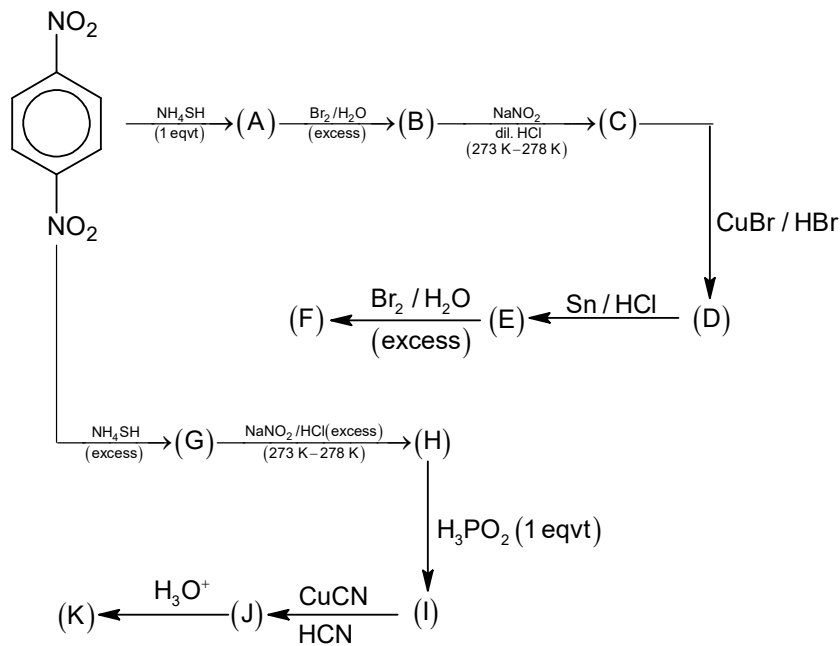
$$\Delta G^\circ = 155880 - 100192.35$$

$$= 55,687.65 \text{ J}$$

$$\Delta G^\circ = 55.68 \text{ kJ}$$

Question Stem for Question Nos. 35 and 36

Question Stem



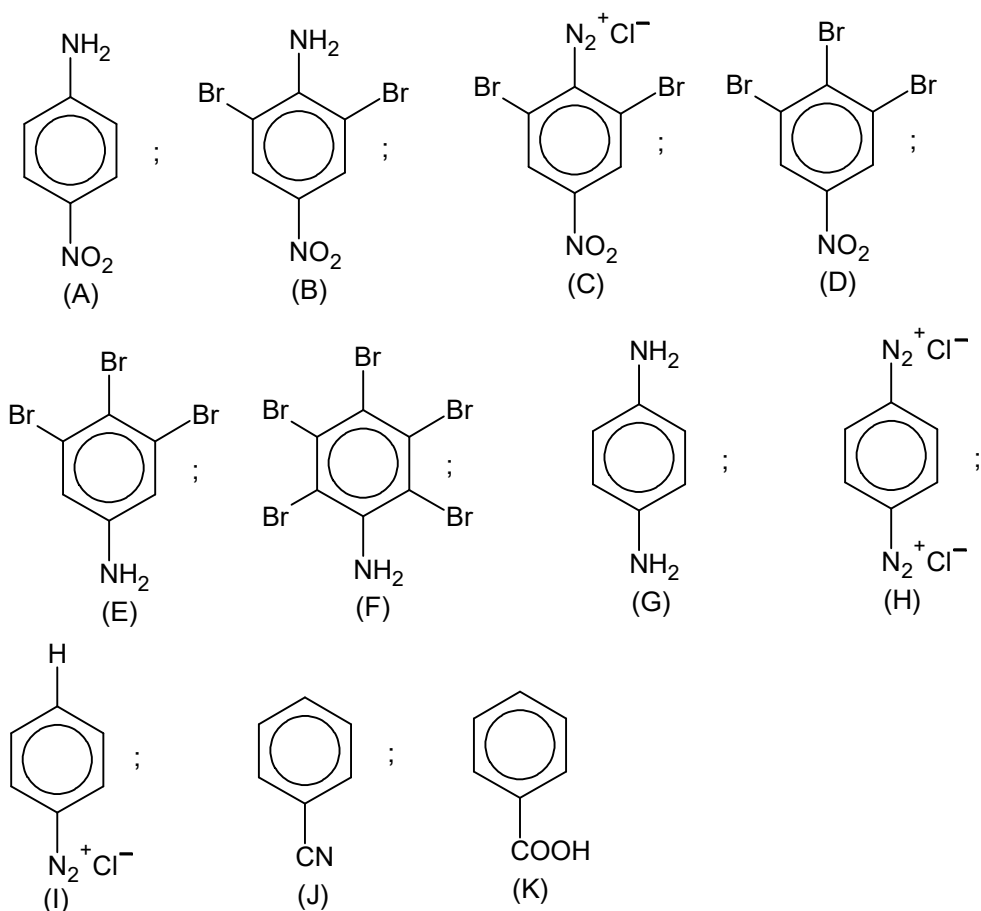
[Atomic masses : C = 12; N = 14; Br = 80; H = 1; Cl = 35.5]

Ratio of the number of halogen atoms present in a molecule of (B) to that of the number of halogen atoms present in a molecule of (F) is 'x' and the ratio of molecular weights of 'D' to that of (K) is 'y'.

35. The value of 'x' is _____

Ans. 0.40

Sol.



36. The value of 'y' is _____

Ans. 2.95
Range (2.94 – 2.96)

Sol. Molar mass of (D) = 360
Molar mass of (K) = 122

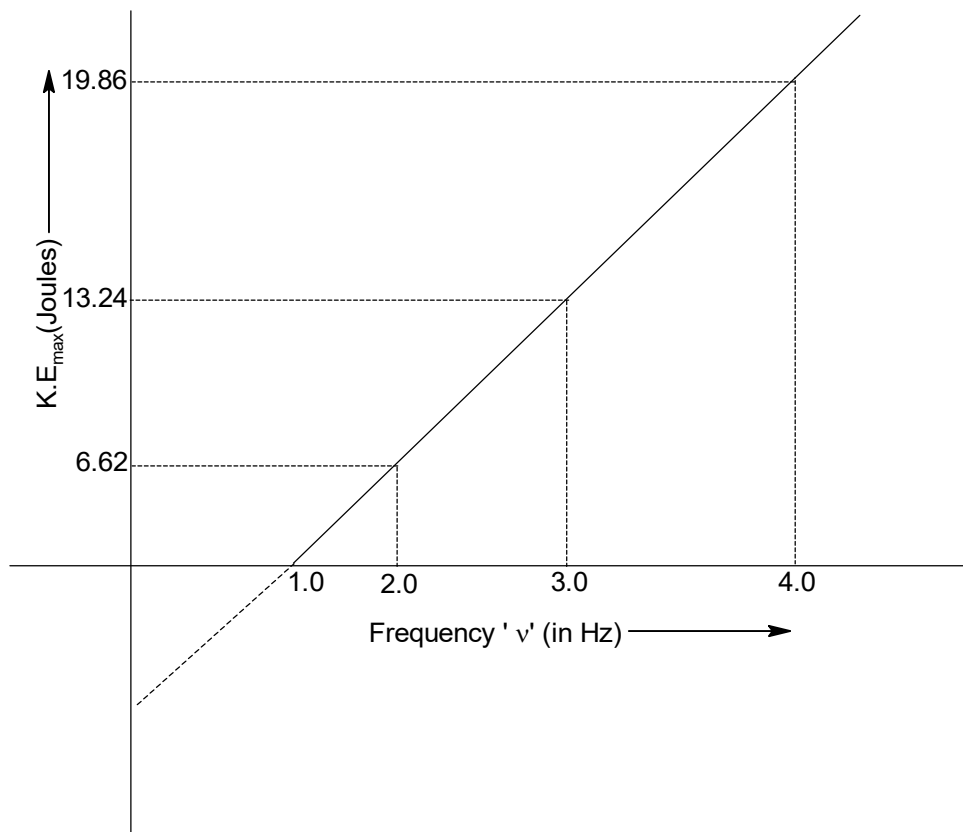
So, ratio of molar mass of (D) to that of (K) = $\frac{360}{122} = 2.95$

Question Stem for Question Nos. 37 and 38

Question Stem

Consider the given plot of $KE_{\max}$ against frequency (ν) of incident light.

(Graph not to scale)



Here, frequency ' ν ' is to be multiplied by 10^{15} and maximum kinetic energy ($KE_{\max}$) to be multiplied by 10^{-19} .

The work function of the metal be ' x ' eV.

Planck's constant (h) = 6.62×10^{-34} J.s and speed of light (c) = 3×10^8 m/sec.

Let, electronic charge (e) = 1.6×10^{-19} Coulomb.

If a light radiation of wavelength 100 nm is allowed to strike surface of the metal (taken in the experiment for which above graph has been plotted), then, maximum speed of photoelectron ejected is found to be $y \times 10^6$ m/s.

37. The value of ' x ' is _____

Ans. Range (4.11 – 4.15)

Sol. $\phi = h\nu_0$
 $= 6.62 \times 10^{-34} \times 1 \times 10^{15}$ J
 $= 6.62 \times 10^{-19}$ J

$$= \frac{6.62 \times 10^{-19}}{1.6 \times 10^{-19}} = 4.1375 \text{ eV}$$

38. The value of 'y' is _____

Ans. Range (1.70 – 1.72)

Sol. Energy of photon $\frac{hc}{\lambda} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{1000 \times 10^{-10}} = 3 \times 6.62 \times 10^{-19} \text{ J}$

Work function (ϕ) = $6.62 \times 10^{-19} \text{ J}$

$\therefore KE_{\max} = 3 \times 6.62 \times 10^{-19} - 6.62 \times 10^{-19}$

$KE_{\max} = 2 \times 6.62 \times 10^{-19} \text{ J} = \frac{1}{2} \times m \times V_{\max}^2$

$V_{\max} = \sqrt{\frac{2 \times 2 \times 6.62 \times 10^{-19}}{9.1 \times 10^{-31}}} = 1.7 \times 10^6 \text{ m/s}$

Mathematics

PART – III

Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. Let ΔPQR be equilateral with side length a . On side PQ produced a point A is taken such that P lies in between A and Q . If r_1 is in radius of ΔAPR and r_2 be ex-radius of ΔAQR with respect to side QR , then $\frac{r_1 + r_2}{a}$ is equal to

- (A) $\frac{\sqrt{3}}{2}$
 (B) $\sqrt{2}$
 (C) 2
 (D) $\frac{1}{\sqrt{2}}$

Ans. A

Sol. As $M_1M_2 = m_1m_2$

$$M_1M_2 = M_1P + PQ + QM_2 = \frac{r_1}{\sqrt{3}} + a + \frac{r_2}{\sqrt{3}}$$

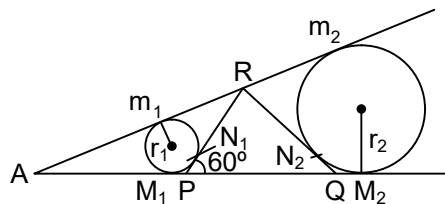
$$m_1m_2 = m_1R + Rm_2 \quad \dots (1)$$

$$= N_1R + N_2R$$

$$= a - \frac{r_1}{\sqrt{3}} + a - \frac{r_2}{\sqrt{3}} \quad \dots (2)$$

From equation (1) and (2), we get $a + \frac{r_1 + r_2}{\sqrt{3}} = 2a - \left(\frac{r_1 + r_2}{\sqrt{3}} \right)$

$$2 \left(\frac{r_1 + r_2}{\sqrt{3}} \right) = a$$



40. A person has 6 cards A, K, 2, Q, 10, 9. The person randomly draws the cards one by one with replacement till he gets 3 consecutive A. If P_n represents the probability that atleast n cards are drawn, then $216P_n - 5P_{n-3} - 30P_{n-2} - 180P_{n-1}$ is (For $n > 3$)

- (A) 0
 (B) 1
 (C) 2
 (D) 3

Ans. A

Sol.
$$P_n = \frac{1}{6} \left\{ \frac{5}{6} \cdot \frac{1}{6} \cdot P_{n-3} + \frac{5}{6} \cdot P_{n-2} \right\} + P_{n-1} \cdot \frac{5}{6}$$

$$216P_n - 5P_{n-3} - 30P_{n-2} - 180P_{n-1} = 0$$

41. Let A and B be 3×3 matrices and $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $B = PAP$, then $\lim_{n \rightarrow \infty} \frac{\text{tr}(A^n + PB^n P)}{\text{tr}(A^n)}$ is equal to
 (where $|A^n| \neq 0$, tr represents trace of square matrix)

- (A) 1
 (B) 2
 (C) 3
 (D) 4

Ans. B

Sol. $B^n = PA^n P$ ($P^2 = I$)
 $\Rightarrow PB^n P = P(PA^n P)P = A^n$

42. Let $f(x) = \frac{1}{\sin^2 x - (x-a)^2}$, $x \in [0, 1]$ and $a \in [2, 3]$, then which of the following is correct?

- (A) $f(x)$ always increases in $(0, 1)$
 (B) $f(x)$ always decreases in interval $(0, 1)$
 (C) there exists a point of local minima in $(0, 1)$
 (D) there exists a point of local maxima in $(0, 1)$

Ans. B

Sol. $f'(x) = -\frac{1}{(\sin^2 x - (x-a)^2)^2} \{ \sin 2x - 2(x-a) \} < 0 \quad \forall x \in (0, 1)$

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. Let the value of $\log_2 \left(\prod_{a=1}^{2015} \prod_{b=1}^{2015} \left(1 + e^{\frac{2\pi i ab}{2015}} \right) \right)$ is N, then which of the following is/are TRUE?

- (A) N is divisible by 5
 (B) N is divisible by 3
 (C) $N > 13000$
 (D) N is divisible by 61

Ans. A, B, C, D

Sol. $N = 13725$ if n is odd, then $\prod_{b=1}^n \left(1 + e^{\frac{2\pi i ab}{n}}\right) = 2^{\gcd(a,n)}$

44. Chords at right angles are drawn through $P(\theta)$ lying on ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ and chords joining extremities meets normal at Q , then which of the following is/are correct?

- (A) Q is same for all such chords drawn through $P(\theta)$
- (B) $y = 0$ is angle bisector of $\angle PCQ$, (where C is centre of ellipse)
- (C) Locus of Q is ellipse for different position of P
- (D) Q is not same for all such chords drawn from P

Ans. A, B, C

Sol. Coordinate of Q are, $\frac{6\cos\theta}{5}, \frac{-3\sin\theta}{5}$

45. Let $f : [1, \infty) \rightarrow \mathbb{R}$ is a differentiable function such that $f'(x) = \frac{1}{x^2 + f^2(x)}$ and $f(1) = 1$, then which of the following is/are correct?

- (A) $f(x)$ is strictly increasing in $(1, \infty)$
- (B) $\lim_{x \rightarrow \infty} f(x) < 2$
- (C) The number of solution of $f(x) = 1 + 2x$ is 0
- (D) $f(3) < 2$

Ans. A, B, C, D

Sol. $f'(x) > 0 \forall x \in (1, \infty)$ so $f(x)$ is strictly increasing

$$\text{So, } f(x) \geq f(1) \forall x \in (1, \infty) \Rightarrow f'(x) \leq \frac{1}{x^2 + 1}$$

$$\text{So, } f(x) = 1 + \int_1^x f'(t) dt \leq 1 + \int_1^x \frac{1}{t^2 + 1} dt \leq 1 + |\tan^{-1} t|_1^x \leq 1 + \tan^{-1} x - \frac{\pi}{4}$$

$$x \rightarrow \infty, f(x) \rightarrow 1 + \frac{\pi}{4}$$

46. If all roots of the equation $x^8 - 4x^7 + b_6x^6 + b_5x^5 + b_4x^4 + b_3x^3 + b_2x^2 + b_1x + \frac{1}{2^8} = 0$ are positive and real, then which of the following is/are correct?

- (A) $\frac{b_2 b_6}{b_4} = 0.7$

(B) $\frac{b_2 b_6}{b_4} = 0.5$

(C) $\frac{b_3 b_5}{b_4} = \frac{14}{5}$

(D) $\frac{b_2}{b_1} = -7$

Ans. A, C, D

Sol. Let α_i for $i = 1, 2, 3, 4, 5, 6, 7, 8$ are positive real roots

$$\sum \alpha_i = 4, \prod \alpha_i = \frac{1}{2^8} \text{ using AM} \geq \text{GM}$$

$$\frac{\sum \alpha_i}{8} \geq \left(\frac{1}{2^8} \right)^{\frac{1}{8}} \text{ as AM = GM} \Rightarrow \text{all roots are equal to } \frac{1}{2}$$

$$\frac{b_2 b_6}{b_4} = \frac{{}^8C_6 \left(\frac{1}{2} \right)^6 \cdot {}^8C_2 \left(\frac{1}{2} \right)^2}{{}^8C_6 \left(\frac{1}{2} \right)^4}$$

47. Let $g(x) = \frac{\cos \pi x}{x(1-x)}$, then which of the following is/are correct?

(A) $g''\left(\frac{1}{2}\right) = 0$

(B) $\int_0^1 g(x) dx = 0$

(C) $g''\left(\frac{1}{2}\right) = -\frac{1}{2}$

(D) $g(x) < 0 \forall x \in \left(\frac{1}{2}, 1\right)$

Ans. A, B, D

Sol. $g\left(\frac{1}{2} - x\right) = -g\left(\frac{1}{2} + x\right) \Rightarrow -g''\left(\frac{1}{2} - x\right) = g''\left(\frac{1}{2} + x\right) \Rightarrow g''\left(\frac{1}{2}\right) = 0$

48. The value of $\sum_{r=0}^{10} {}^{20+r}C_{20} \cdot {}^{20-r}C_{10}$ is equal to

(A) ${}^{39}C_9$

(B) ${}^{41}C_{10}$

(C) ${}^{41}C_{31}$

(D) ${}^{39}C_{30}$

Ans. B, C

Sol. Coefficient of x^{10} in $(1-x)^{-32}$

Section – B (Maximum Marks: 12)

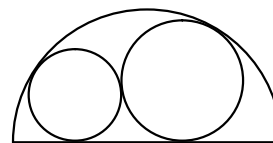
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. The area bounded by region $\{(x, y) : |x + ay + a^2| \leq 1 \forall a \in [0, 1]\}$ is A, then $[A]$ is (where $[.]$ denotes greatest integer function)

Ans. 3

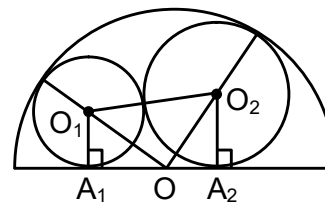
Sol. Required area = $\frac{1}{2} + \int_{-1}^0 (2\sqrt{1+x}) dx + \int_0^1 (-x + 2\sqrt{1+x}) dx + \int_0^1 (-2\sqrt{1+x} + 2+x) dx$
 $= \frac{1}{2} + \frac{4}{3} + 2 = \frac{23}{6}$

50. Two circles with radius r_1 and r_2 touches each other externally and touch a semicircle of radius 1 internally as shown in figure. If maximum value of $r_1 + r_2$ is k, then $\frac{3k}{\sqrt{2}-1}$ is



Ans. 6

Sol. $A_1A_2 = \sqrt{(r_1+r_2)^2 - (r_1-r_2)^2} = 2\sqrt{r_1r_2}$
 and $A_1A_2 = A_1O + OA_2 = \sqrt{(1-r_1)^2 - r_1^2} + \sqrt{(1-r_2)^2 - r_2^2}$
 $= \sqrt{1-2r_1} + \sqrt{1-2r_2}$
 $\Rightarrow \sqrt{1-2r_1} + \sqrt{1-2r_2} = 2\sqrt{r_1r_2}$
 $\Rightarrow r_1 + r_2 = 2\sqrt{r_1r_2}(\sqrt{2} - \sqrt{r_1r_2}) \leq (r_1 + r_2)(\sqrt{2} - \sqrt{r_1r_2})$
 $\sqrt{r_1r_2} \leq \sqrt{2} - 1 \Rightarrow r_1 + r_2 \leq 2(\sqrt{2} - 1)$



51. Let $P(x)$ and $Q(x)$ be polynomials with real coefficients such that $P(x)(x^2 - 3x + 2) = Q(x)(x^2 + 3x + 2)$. Then minimum number of real roots of the equation $P(x) \cdot Q(x) + x^4 - 5x^2 + 4 = 0$ is

Ans. 4

Sol. Let $g(x) = P(x) \cdot Q(x) + x^4 - 5x^2 + y$, the $g(-2) = 0$, $g(2) = 0$, $g(-1) = 0$, $g(1) = 0$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $P(1, s, t)$ be a point and $L_1 : x + y = 1, z + 2y = 2$, $L_2 : x + y = 1, z - 3y = 2$ be straight lines, then

52. If there exist infinite number of lines passing through P and intersecting both the lines, then the value of s is

Ans. 0.00

53. If there exist unique line passing through P and intersecting both the lines L_1 and L_2 , then number of values of s such that P lies on the plane $x + y = 1$ is

Ans. 0.00

Sol. (for Q. 52-53)
Lines L_1 and L_2 are coplanar and lies on plane $x + y = 1$

Question Stem for Question Nos. 54 and 55

Question Stem

Let $P(-1, \alpha)$, $Q(2, \beta)$ be focii of the ellipse which touches $y = x$ at $(1, 1)$. If length of major axis of ellipse is $\frac{3}{2}\sqrt{17}$, then

54. The value of $\alpha(> 0)$ is

Ans. 0.50

55. The value of ordinate of centre of ellipse if $(\alpha > 0)$ is

Ans. 2.75

Sol. (for Q. 54-55)
Image of one focus about tangent, point of contact of tangent and other focus are collinear

Question Stem for Question Nos. 56 and 57

Question Stem

Let $y = f(x)$ satisfy the differential equation $(x + 2)(x + y + 2)dy = y^2dx$, for $x > 0$ and $f(1) = 6$, then

56. Number of points of intersection of $y = f(x)$ and $y = x + 2$ is

Ans. 1.00

57. Number of points of intersection of $y = f(x)$ and $y = (x + 2)^2$ is

Ans. 0.00

Sol. $(x + 2)^2 dy + (x + 2)y dy = y^2 dx$
 $\frac{1}{y} dy + \frac{1}{(x + 2)} dy - \frac{y}{(x + 2)^2} dx = 0$
 $\frac{1}{y} dy + d\left(\frac{y}{x + 2}\right) = 0 \Rightarrow \ln y + \frac{y}{x + 2} = c$